

MIRNY, Russian Federation

WMO: 247260

Lat: 62.535N

Lon: 114.039E

Elev: 352

StdP: 97.17

Time Zone: 9.00 (E09)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-45.0	-42.7	-47.8	0.0	-44.9	-45.1	0.0	-42.5	8.8	-18.1	8.0	-20.5	1.0	240	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.0	18.0	26.5	17.0	24.2	16.1	19.5	25.5	18.3	24.3	17.1	22.7	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.9	21.1	16.1	11.9	20.4	15.0	11.1	19.2	56.9	25.7	52.9	23.9	49.4	22.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.4	DB	-45.9	32.3	3.3	1.6	-48.3	33.4	-50.2	34.4	-52.0	35.3	-54.4	36.5	(4)
(5)			WB	-45.9	21.4	3.2	1.5	-48.2	22.5	-50.1	23.4	-51.9	24.3	-54.2	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.3	-29.5	-26.7	-14.9	-4.2	5.3	15.0	17.4	13.7	5.1	-6.4	-21.5	-29.9
	DBStd	18.19	7.66	7.40	7.45	5.63	4.83	4.83	3.91	3.68	4.45	6.31	8.50	8.33
	HDD10.0	6477	1224	1028	772	426	159	12	1	11	155	509	944	1236
	HDD18.3	9041	1482	1262	1030	676	403	120	65	150	396	768	1194	1494
	CDD10.0	539	0	0	0	0	14	162	229	124	9	0	0	0
	CDD18.3	60	0	0	0	0	0	21	34	5	0	0	0	0
	CDH23.3	678	0	0	0	0	5	251	356	66	1	0	0	0
	CDH26.7	183	0	0	0	0	1	66	105	12	0	0	0	0
Wind	WSAvg	3.2	2.7	2.9	3.1	3.5	3.8	3.4	3.2	3.1	3.3	3.4	2.8	2.6
Precipitation	PrecAvg	378	15	12	15	20	32	50	52	55	39	40	28	20
	PrecMax	510	31	29	27	43	53	93	95	92	70	69	48	93
	PrecMin	284	6	6	3	6	14	18	18	19	5	12	10	5
	PrecStd	50	6	5	6	9	11	18	19	20	17	15	9	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.8	-5.2	5.1	12.0	22.9	31.1	32.0	28.6	21.2	8.2	-0.1	-5.1
		MCWB	-9.0	-5.7	0.6	5.2	12.4	18.7	20.0	17.6	12.8	4.8	-0.4	-5.4
	2%	DB	-13.0	-9.3	1.1	9.0	19.3	28.8	29.9	25.0	17.1	5.1	-2.0	-10.0
		MCWB	-13.2	-9.7	-1.5	3.3	10.3	17.3	19.0	16.0	10.4	3.0	-2.5	-10.0
	5%	DB	-15.9	-12.2	-1.9	6.2	16.2	26.2	27.1	22.5	14.2	3.0	-4.4	-14.1
		MCWB	-16.1	-12.7	-3.6	1.5	9.0	16.4	17.9	15.0	9.3	1.8	-4.7	-14.2
	10%	DB	-18.9	-15.6	-4.1	4.1	13.9	24.0	25.0	20.2	11.9	1.1	-8.8	-17.9
		MCWB	-19.0	-15.8	-5.6	0.4	7.6	15.3	17.0	14.2	8.3	0.2	-9.1	-18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.0	-5.7	1.2	5.4	13.4	20.5	21.7	19.1	13.6	5.2	-0.6	-5.8
		MCDB	-8.4	-5.2	4.0	10.7	20.5	26.7	28.1	25.0	18.7	7.0	-0.5	-5.5
	2%	WB	-13.3	-9.9	-1.6	3.5	11.2	18.6	20.1	17.3	11.7	3.5	-2.5	-10.6
		MCDB	-13.1	-9.6	0.6	8.0	17.7	25.6	26.0	22.3	15.3	4.8	-2.2	-10.6
	5%	WB	-16.1	-12.9	-3.5	2.2	9.5	17.3	19.0	16.2	10.1	1.9	-4.9	-14.7
		MCDB	-16.0	-12.4	-1.8	5.6	15.8	24.5	25.0	20.6	13.1	2.9	-4.6	-14.7
	10%	WB	-19.1	-15.9	-5.6	0.8	8.0	16.2	18.0	15.1	8.8	0.3	-9.2	-18.2
		MCDB	-19.0	-15.7	-4.2	3.7	13.2	22.3	23.6	19.1	11.7	1.0	-8.9	-18.1
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.9	10.5	10.2	9.6	10.9	10.2	9.9	8.0	6.3	6.6	6.2
		MCDBR	8.5	8.1	10.7	11.4	13.8	14.2	13.8	13.5	12.1	7.2	8.6	8.7
		MCWBR	8.4	7.7	8.7	7.5	7.8	6.2	5.9	6.4	7.0	5.5	8.7	8.9
	5% WB	MCDBR	8.5	8.1	10.2	9.9	12.8	12.5	11.8	11.3	10.4	6.7	8.6	9.0
		MCWBR	8.4	7.7	8.4	7.1	7.9	6.4	6.3	6.3	6.8	5.5	8.7	9.0
Clear-Sky Solar Irradiance	taub	0.219	0.261	0.293	0.326	0.354	0.399	0.435	0.422	0.320	0.276	0.213	0.190	
	taud	1.929	2.053	2.080	2.124	2.318	2.310	2.195	2.180	2.477	2.316	1.962	1.873	
	Ebn at Noon	416	677	800	852	861	823	773	744	790	676	417	249	
	Edh at Noon	37	71	102	122	111	115	125	117	71	55	34	20	
All-Sky Solar Radiation	RadAvg	0.31	1.24	2.95	4.87	5.44	5.74	5.20	3.99	2.41	1.13	0.41	0.14	
	RadStd	0.02	0.07	0.12	0.24	0.34	0.37	0.35	0.24	0.24	0.09	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page